

Advanced Customer Segmentation

E-Commerce Behavioral Analytics & Machine Learning

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1. Project Overview

Business Problem

E-commerce companies struggle to identify valuable customers and understand behavioral patterns from large transaction volumes, leading to inefficient marketing.

Objective

Develop a framework using RFM, PCA, K-Means, and DBSCAN to identify customer personas and generate actionable business intelligence.

2. Project Architecture

The solution follows an end-to-end analytical pipeline transforming raw data into intelligence.

PROJECT ARCHITECTURE DIAGRAM



3. Data & Feature Engineering

RFM Analysis

Feature	Description
Recency	Days since last purchase
Frequency	Total number of purchases
Monetary	Total customer lifetime value

Behavioral Metrics

Enhanced insights beyond standard RFM include Avg. Quantity, Preferred Payment methods, and Discount Sensitivity (influence ~10%).

4. Machine Learning Methodology

PCA

Dimensionality reduction to eliminate redundancy and improve clustering efficiency.

K-Means

Primary segmentation identifying 3 core customer personas.

DBSCAN

Utilized for outlier detection and validating traditional clustering results.

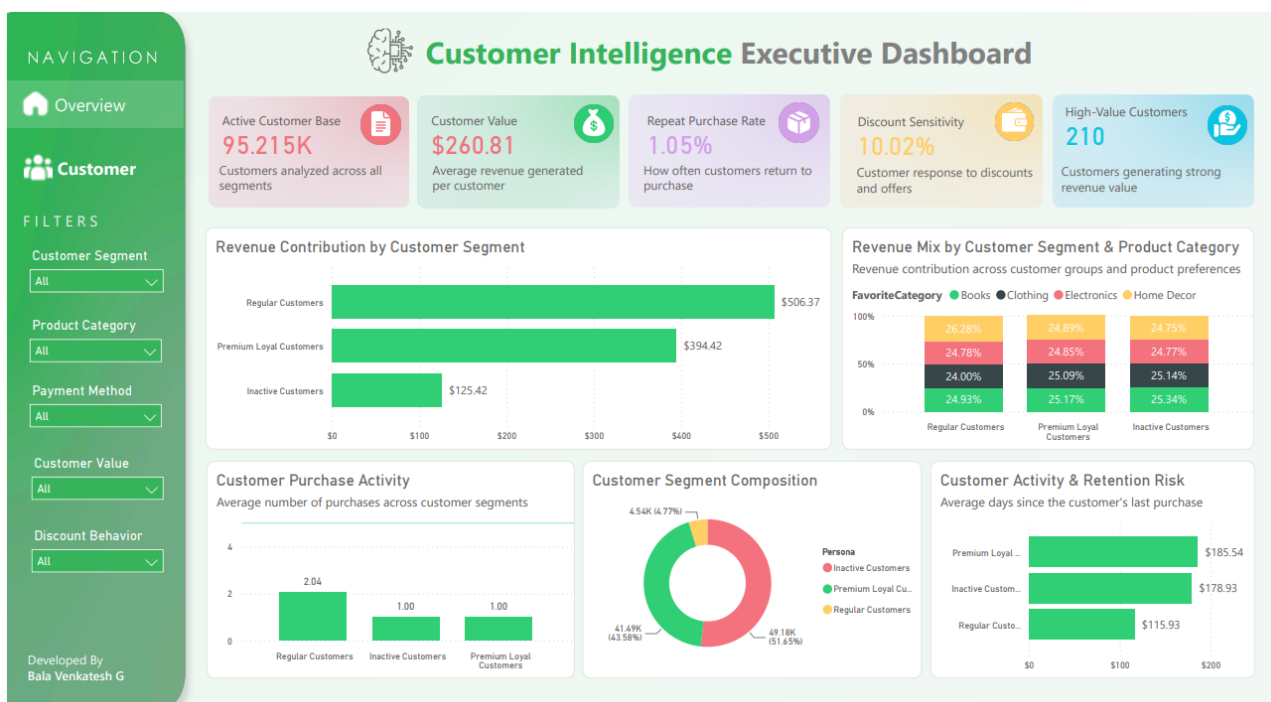
5. Customer Personas

Premium Loyal Customers: High value, consistent spending. Action: Priority rewards and exclusive previews.

Regular Customers: Largest segment, highest total revenue contribution. Action: Retention programs and upselling.

Inactive Customers: Low engagement, high churn risk. Action: Win-back campaigns and personalized discounts.

6. Executive Dashboard



7. Behavioral Intelligence

